

SCQ (Single Correct Type) :

- Complete set of real values of 'a' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real :
 (A) $\left[\frac{3}{4}, \infty\right)$ (B) $[1, \infty)$ (C) $[2, \infty)$ (D) $[0, \infty)$
- Let a, b, c be the lengths of the sides of a triangle (no two of them are equal) and $k \in \mathbb{R}$. If the roots of the equation $x^2 + 2(a + b + c)x + 6k(ab + bc + ca) = 0$ are real, then:
 (A) $k < \frac{2}{3}$ (B) $k > \frac{2}{3}$ (C) $k > 1$ (D) $k < \frac{1}{4}$
- If $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$ are roots of the cubic equation $f(x) = 0$ where α, β, γ are the roots of the cubic equation $3x^3 - 2x + 5 = 0$, then the number of negative real roots of the equation $f(x) = 0$ is:
 (A) 0 (B) 1 (C) 2 (D) 3
- Range of the function $f(x) = \log_2 \left(\frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$ is :
 (A) $(0, \infty)$ (B) $\left[\frac{1}{2}, 1\right]$ (C) $[1, 2]$ (D) $\left[\frac{1}{4}, 1\right]$
- If $[x]^2 - 7[x] + 10 < 0$ and $4[y]^2 - 16[y] + 7 < 0$, then $[x + y]$ cannot be ([.] denotes greatest integer function) :
 (A) 7 (B) 8 (C) 9 (D) Both (B) and (C)
- The function $f(x)$ satisfy the equation $f(1-x) + 2f(x) = 3x \quad \forall x \in \mathbb{R}$, then $f(0) =$
 (A) -2 (B) -1 (C) 0 (D) 1
- If a, b, c are in GP, $a - b, c - a, b - c$ are in HP, then the value of $a + 4b + c$ is
 (A) 12 (B) 0 (C) 11 (D) 2
- Find the value of $\frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$ upto ∞ terms :
 (A) 2 (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{4}$

MCQ (One or more than one correct) :

- Let $\alpha, \beta, \gamma, \delta$ are roots of $x^4 - 12x^3 + \lambda x^2 - 54x + 14 = 0$. If $\alpha + \beta = \gamma + \delta$, then
 (A) $\lambda = 45$ (B) $\lambda = -45$
 (C) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2$ then $\frac{\alpha\beta}{\gamma\delta} = \frac{7}{2}$ (D) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2 \Rightarrow \frac{\alpha\beta}{\gamma\delta} = \frac{2}{7}$

10. If the equation $\cot^4 x - 2 \operatorname{cosec}^2 x + a^2 = 0$ has at least one solution then possible integral values of a can be :
- (A) -1 (B) 0 (C) 1 (D) 2
11. Consider the function $f(x) = \ln((\cos x)^{\cos x} + 1)$; $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (A) $f(x)$ is an even function (B) Range of $f(x)$ is $(\ln(1 + e^{1/e}), \ln 2)$
- (C) Range of $f(x)$ is $\left(\ln\left(1 + \left(\frac{1}{e}\right)^{\frac{1}{e}}\right), \ln 2\right)$ (D) $f(x)$ is neither an even nor an odd function
12. If numbers of ordered pairs (p, q) from the set $S = \{1, 2, 3, 4, 5\}$ such that the function $f(x) = \frac{x^3}{3} + \frac{p}{2}x^2 + qx + 10$ defined from $\mathbb{R}$ to $\mathbb{R}$ is injective, is n then n is divisible by
- (A) 3 (B) 5 (C) 7 (D) 11
13. Let $x, y, z \in \left(0, \frac{\pi}{2}\right)$ are first three consecutive terms of an arithmetic progression such that $\cos x + \cos y + \cos z = 1$ and $\sin x + \sin y + \sin z = \frac{1}{\sqrt{2}}$, then which of the following is/are correct ?
- (A) $\cot y = \sqrt{2}$ (B) $\cos(x - y) = \frac{\sqrt{3} - \sqrt{2}}{2\sqrt{2}}$
- (C) $\tan 2y = \frac{2\sqrt{2}}{3}$ (D) $\sin(x - y) + \sin(y - z) = 0$

Numerical Based Questions :

14. $f(x) = \log_4 \left(\frac{2x^2 - 3x + a}{2x^2 + 3x + 3} \right)$ has range $[\alpha, \beta]$ such that $\alpha + \beta = 0$, then find 'a'.
15. Find the number of integers satisfying the inequality $\sqrt{\log_{1/2} x + 4 \log_2 \sqrt{x}} < \sqrt{2} (4 - \log_{16} x^4)$.
16. If three angles A, B, C are such that $\cos A + \cos B + \cos C = 0$ and if $\lambda \cos A \cos B \cos C = \cos 3A + \cos 3B + \cos 3C$, then value of λ is :
17. Let $f(x) = x^2 - bx + c$, b is an odd positive integer. Given that $f(x) = 0$ has two prime numbers as roots and $b + c = 35$. If the least value of $f(x) \forall x \in \mathbb{R}$ is λ , then $\left\lceil \frac{\lambda}{3} \right\rceil$ is equal to
- (where $[\cdot]$ denotes greatest integer function)
18. Consider the equation $(x^2 + x + 1)^2 - (m - 3)(x^2 + x + 1) + m = 0$, where m is a real parameter. The number of positive integral values of m for which equation has two real roots, is :

19. If the equation $(m^2 - 12)x^4 - 8x^2 - 4 = 0$ has no real roots, then the largest value of m is $p\sqrt{q}$ where p, q are coprime natural numbers, then $p + q =$

Subjective Type Problems :

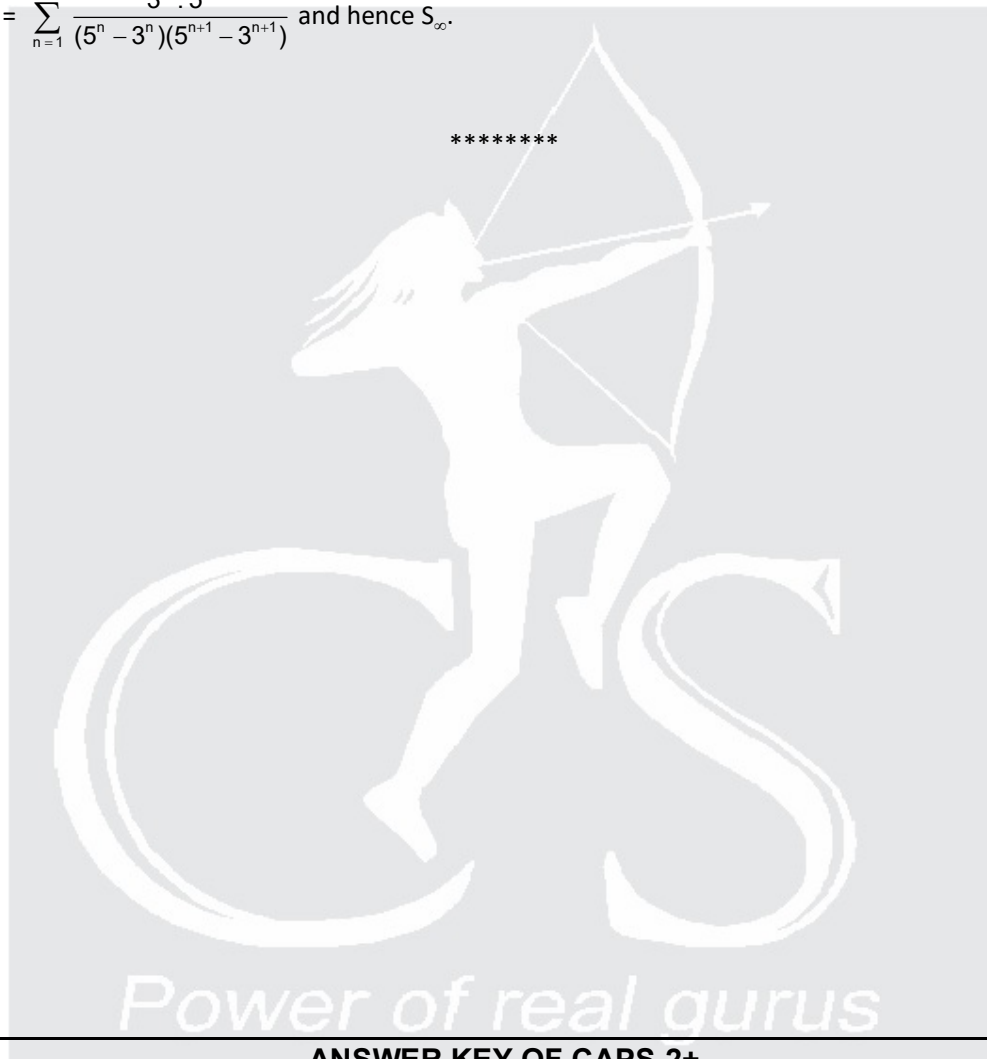
20. Solve the following system of simultaneous equations for x and y :

$$4^{\sin x} + 3^{1/\cos y} = 11$$

$$5.16^{\sin x} - 2.3^{1/\cos y} = 2$$

21. Find the value of

$$S_n = \sum_{n=1}^n \frac{3^n \cdot 5^n}{(5^n - 3^n)(5^{n+1} - 3^{n+1})} \text{ and hence } S_{\infty}.$$



ANSWER KEY OF CAPS-2+

1.	(A)	2.	(A)	3.	(B)	4.	(B)	5.	(C)	6.	(B)	7.	(B)
8.	(C)	9.	(AC)	10.	(ABC)	11.	(AC)	12.	(AB)	13.	(AB)	14.	3
15.	3	16.	12	17.	6	18.	1	19.	5	21.	$\frac{3}{4}$		